

**KEM PORT, Russian Federation**

WMO: 225200

Lat: **64.983N**Lon: **34.800E**Elev: **8**StdP: **101.23**Time Zone: **3.00 (E03)**Period: **06-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -25.1      | -22.0 | -27.8                         | 0.3 | -24.9 | -24.3 | 0.4 | -22.0 | 10.6                  | -5.9 | 9.8 | -6.3 | 1.7                   | 260  | 0.710 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 5.8                    | 24.2            | 17.3 | 21.9 | 16.4 | 20.1 | 15.7 | 18.5                | 22.5 | 17.4 | 20.8 | 16.3 | 19.4 | 3.7                  | 270  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.9                            | 12.1 | 20.3 | 15.9 | 11.3 | 19.2 | 14.8 | 10.5 | 18.2 | 52.3          | 22.7 | 49.1 | 20.7 | 45.8 | 19.5 | 21.9              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |            |            | Extreme Annual Temperature |             |                    |            | n-Year Return Period Values of Extreme Temperature |             |              |             |              |             |              |             |    |    |
|-------------------|------------|------------|----------------------------|-------------|--------------------|------------|----------------------------------------------------|-------------|--------------|-------------|--------------|-------------|--------------|-------------|----|----|
|                   |            |            | Mean                       |             | Standard Deviation |            | n=5 years                                          |             | n=10 years   |             | n=20 years   |             | n=50 years   |             |    |    |
| 1%                | 2.5%       | 5%         | Min                        | Max         | Min                | Max        | Min                                                | Max         | Min          | Max         | Min          | Max         | Min          | Max         | DB | WB |
| (a)               | (b)        | (c)        | (e)                        | (f)         | (g)                | (h)        | (i)                                                | (j)         | (k)          | (l)         | (m)          | (n)         | (o)          | (p)         |    |    |
| (4) <b>9.4</b>    | <b>8.2</b> | <b>7.3</b> | <b>-27.4</b>               | <b>27.6</b> | <b>3.6</b>         | <b>2.6</b> | <b>-30.0</b>                                       | <b>29.5</b> | <b>-32.0</b> | <b>31.0</b> | <b>-34.1</b> | <b>32.4</b> | <b>-36.7</b> | <b>34.3</b> |    |    |
| (5)               |            |            | <b>-27.5</b>               | <b>19.8</b> | <b>3.5</b>         | <b>1.3</b> | <b>-30.1</b>                                       | <b>20.8</b> | <b>-32.2</b> | <b>21.5</b> | <b>-34.1</b> | <b>22.2</b> | <b>-36.7</b> | <b>23.2</b> |    |    |

**Monthly Climatic Design Conditions**

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 2.6    | -9.3  | -8.9  | -4.6  | 0.6   | 6.4   | 11.0  | 14.8  | 13.9  | 9.7   | 3.4   | -1.5  | -5.2 |
|                                                                                  | DBStd       | 9.45   | 6.53  | 6.62  | 5.18  | 3.67  | 4.33  | 3.77  | 3.59  | 3.53  | 3.19  | 3.30  | 3.63  | 5.81 |
|                                                                                  | HDD10.0     | 3100   | 598   | 530   | 454   | 281   | 129   | 32    | 4     | 6     | 43    | 205   | 347   | 472  |
|                                                                                  | HDD18.3     | 5767   | 857   | 763   | 712   | 531   | 370   | 222   | 119   | 144   | 259   | 463   | 597   | 730  |
|                                                                                  | CDD10.0     | 394    | 0     | 0     | 0     | 0     | 17    | 62    | 152   | 127   | 35    | 1     | 0     | 0    |
|                                                                                  | CDD18.3     | 18     | 0     | 0     | 0     | 0     | 0     | 2     | 10    | 7     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 102    | 0     | 0     | 0     | 0     | 7     | 12    | 54    | 30    | 0     | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 17     | 0     | 0     | 0     | 0     | 1     | 2     | 11    | 2     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 3.7    | 3.8   | 3.7   | 3.8   | 3.6   | 3.7   | 3.5   | 3.2   | 3.4   | 3.6   | 4.0   | 4.1   | 4.0  |
| Precipitation                                                                    | PrecAvg     | 434    | 20    | 15    | 16    | 24    | 40    | 56    | 46    | 60    | 54    | 49    | 37    | 26   |
|                                                                                  | PrecMax     | 708    | 42    | 35    | 36    | 47    | 97    | 122   | 110   | 113   | 109   | 84    | 59    | 69   |
|                                                                                  | PrecMin     | 274    | 6     | 3     | 3     | 1     | 7     | 10    | 2     | 15    | 18    | 6     | 10    | 2    |
|                                                                                  | PrecStd     | 88     | 10    | 9     | 8     | 12    | 22    | 32    | 27    | 22    | 30    | 20    | 13    | 16   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 2.5   | 4.2   | 7.6   | 14.2  | 23.6  | 25.0  | 28.3  | 26.5  | 20.3  | 12.4  | 7.2   | 4.5  |
|                                                                                  |             | MCWB   | 1.6   | 2.3   | 3.4   | 8.4   | 13.2  | 16.5  | 19.1  | 18.5  | 15.0  | 10.0  | 6.2   | 3.1  |
|                                                                                  | 2%          | DB     | 1.3   | 2.3   | 4.7   | 9.6   | 18.5  | 21.3  | 24.4  | 23.6  | 17.8  | 10.3  | 5.3   | 3.0  |
|                                                                                  |             | MCWB   | 0.5   | 1.1   | 1.7   | 5.2   | 11.1  | 14.5  | 18.1  | 17.1  | 14.1  | 8.5   | 4.5   | 1.9  |
|                                                                                  | 5%          | DB     | 0.2   | 1.2   | 3.2   | 7.4   | 16.0  | 18.8  | 22.1  | 21.5  | 16.0  | 8.9   | 3.7   | 2.1  |
|                                                                                  |             | MCWB   | -0.3  | 0.4   | 1.0   | 3.7   | 10.0  | 13.4  | 17.1  | 16.6  | 12.9  | 7.6   | 2.9   | 1.2  |
|                                                                                  | 10%         | DB     | -0.9  | -0.4  | 2.0   | 5.8   | 13.6  | 16.9  | 20.3  | 19.6  | 14.4  | 7.8   | 2.7   | 1.2  |
|                                                                                  |             | MCWB   | -1.3  | -1.1  | 0.4   | 2.8   | 9.0   | 12.7  | 16.4  | 15.9  | 12.1  | 6.6   | 2.0   | 0.6  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 1.6   | 2.4   | 3.6   | 8.8   | 14.0  | 18.3  | 20.6  | 19.2  | 15.8  | 10.6  | 6.3   | 3.6  |
|                                                                                  |             | MCDB   | 2.2   | 3.6   | 7.0   | 14.0  | 19.3  | 23.6  | 25.2  | 24.2  | 18.7  | 11.9  | 7.1   | 4.1  |
|                                                                                  | 2%          | WB     | 0.6   | 1.2   | 2.2   | 5.4   | 12.1  | 15.5  | 19.0  | 18.2  | 14.4  | 9.0   | 4.5   | 2.1  |
|                                                                                  |             | MCDB   | 1.2   | 2.0   | 4.1   | 8.7   | 17.2  | 19.5  | 22.9  | 22.3  | 16.9  | 9.7   | 5.1   | 2.8  |
|                                                                                  | 5%          | WB     | -0.3  | 0.4   | 1.3   | 4.3   | 10.6  | 14.2  | 17.8  | 17.2  | 13.4  | 7.6   | 3.0   | 1.3  |
|                                                                                  |             | MCDB   | 0.2   | 1.2   | 2.8   | 6.8   | 15.3  | 17.7  | 20.9  | 20.6  | 15.6  | 8.6   | 3.5   | 2.0  |
|                                                                                  | 10%         | WB     | -1.3  | -1.0  | 0.6   | 3.2   | 9.3   | 13.2  | 16.8  | 16.1  | 12.4  | 6.7   | 2.1   | 0.6  |
|                                                                                  |             | MCDB   | -0.9  | -0.4  | 1.9   | 5.4   | 13.1  | 16.3  | 19.7  | 19.1  | 14.2  | 7.8   | 2.6   | 1.1  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 5.4   | 5.7   | 6.8   | 6.5   | 6.6   | 6.2   | 5.8   | 5.8   | 5.4   | 3.9   | 3.7   | 4.4  |
|                                                                                  |             | MCDBR  | 4.4   | 4.7   | 6.4   | 8.7   | 11.7  | 10.2  | 9.7   | 9.1   | 8.0   | 5.3   | 4.0   | 4.0  |
|                                                                                  |             | MCWBR  | 4.4   | 4.4   | 4.6   | 5.3   | 6.1   | 5.2   | 4.9   | 4.2   | 4.8   | 4.0   | 3.7   | 3.7  |
|                                                                                  | 5% WB       | MCDBR  | 4.1   | 4.9   | 5.6   | 8.0   | 10.6  | 8.2   | 8.2   | 7.8   | 6.7   | 4.4   | 3.9   | 3.7  |
|                                                                                  |             | MCWBR  | 4.1   | 4.6   | 4.4   | 5.2   | 6.5   | 4.9   | 4.8   | 4.4   | 4.2   | 3.9   | 3.7   | 3.6  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.219  | 0.280 | 0.309 | 0.342 | 0.353 | 0.349 | 0.378 | 0.355 | 0.331 | 0.303 | 0.345 | 0.356 |      |
|                                                                                  | taud        | 1.957  | 2.094 | 2.154 | 2.275 | 2.425 | 2.487 | 2.436 | 2.476 | 2.521 | 2.363 | 2.560 | 2.606 |      |
|                                                                                  | Ebn at Noon | 280    | 591   | 755   | 825   | 854   | 867   | 824   | 811   | 756   | 587   | 273   | 116   |      |
|                                                                                  | Edh at Noon | 23     | 61    | 90    | 102   | 97    | 94    | 96    | 84    | 65    | 47    | 21    | 12    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.12   | 0.69  | 2.24  | 3.74  | 4.67  | 5.47  | 5.01  | 3.57  | 2.02  | 0.73  | 0.16  | 0.03  |      |
|                                                                                  | RadStd      | 0.01   | 0.11  | 0.23  | 0.36  | 0.40  | 0.48  | 0.55  | 0.37  | 0.21  | 0.10  | 0.02  | 0.00  |      |

**Historical Trends**

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (8 neighbors) | +0.62 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -227    | N/A     | N/A     |

Nomenclature: See separate page